HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Performed RF measurements using **spectrum analysers** and **vector network analysers (VNAs)**, documenting results for **wireless product verification**.
- Assisted with **RF design validation** and **performance characterisation** of communication systems, contributing to the development of **life safety products**.
- Supported **antenna testing**, matching, and RF performance optimisation by conducting **conducted and radiated measurements**.
- Executed **PCB soldering and rework**, including modification of **small surface-mount components** and **matching-network updates**.
- Investigated unexpected measurement results and contributed to **root-cause analysis** for RF circuitry issues.
- Prepared **technical reports** and maintained detailed laboratory records of **engineering findings** and test configurations.
- Collaborated with Hardware, Firmware, and Validation teams to support **RF engineering activities** throughout the product development lifecycle.

SKILLS

RF Engineering – RF measurements, Antenna tuning, Wireless communication systems, RF design validation, Performance characterisation, Regulatory compliance, Conducted measurements, Radiated measurements, RF PCB layout, Impedance matching, Antenna matching techniques

Test Equipment – Spectrum Analysers, Vector Network Analysers (VNAs), RF Signal Generators, RF Power Meters, Oscilloscopes

PCB Rework – Soldering, Surface-mount components (0603, 0402), PCB modification, Component replacement, Matching-network updates

Software & Tools – Linux, Git, Jira, Confluence, Python scripting

Core Technologies – Java Spring Boot, REST APIs, SQL (PostgreSQL, MySQL, Oracle), AWS (EC2, S3, Lambda, CloudWatch), Docker

PROJECTS

RF System Measurement and Validation

- Configured and operated **spectrum analysers** and **VNAs** to perform detailed RF performance characterisation for new wireless modules.
- Executed **conducted and radiated tests** on prototype devices, documenting findings and identifying areas for **antenna tuning** and optimisation.
- Contributed to the preparation of technical documentation, detailing test methodologies and results for **regulatory assessment** support.

PCB Modification for RF Performance Tuning

- Performed precise **PCB soldering and rework** on development boards to modify RF **matching networks** under magnification.
- Replaced small **surface-mount components** (0603 packages) to adjust impedance and improve signal integrity for wireless transceivers.
- Verified the impact of PCB modifications through subsequent RF measurements, ensuring repeatable **antenna matching** improvements.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)